Shruti Swaminathan

5th year BS-MS (Biology) student at IISER Mohali, India
MS Thesis Intern at Friedrich Miescher Laboratory of the Max Planck Society, Tübingen
DST-INSPIRE Scholar

ms22139@iisermohali.ac.in
shruti.swaminathan20@gmail.com
(+91) 9527067256
LinkedIn

Profile

5th year BS-MS *(Dual Degree)* student at Indian Institute of Science Education and Research (IISER) Mohali majoring in Biological Sciences with a minor in Data Science. I also have a keen interest in Science Education and Science Communication.

Education

Indian Institute of Science Education and Research Mohali **2022 – 2027**
Integrated BS-MS Degree – ***CPI: 9.74/10*** *Mohali, India*
A 5-year bachelor's and master's degree with a **major in Biological Sciences and a minor in Data Science**

Elpro International School **2020 – 2022**
CBSE Class XII national-level standardized examination — ***96%*** *Pune, India*
Senior Secondary Education (Grades 11–12), Science stream (Physics, Chemistry, Mathematics, Biology, English)

Wisdom World School, Wakad **2017 – 2020**
ICSE Class X national-level standardized examination — ***98%*** *Pune, India*
Secondary Education (Grades 8–10)

Experience

Diffusiophoresis in self-organizing systems *in vitro* **June 2026 – present**
Masters Thesis under Dr. Beatrice Ramm *Friedrich Miescher Laboratory, MPI for Biology Tübingen*
Ramm Lab

- Selected under the IISER-MPG program to conduct the research for my Masters thesis at the Max Planck Institute for Biology in Tübingen, Germany
- Looking at the effects of diffusiophoretic transport in the two self-organizing protein systems in vitro and potential coupling of the systems via diffusiophoresis
- Protein purification using Ni-NTA chromatography, Preparation of single unilamellar vesicles (SUVs) and 2D supported lipid bilayers (SLBs) of varying lipid compositions, Confocal microscopy

Microbial evolution through the lens of phages **May 2025 – July 2025**
Summer Internship under Dr. Yashraj Chavhan *IISER Thiruvananthapuram*
Chavhan Lab for Understanding Evolution (CLUE)

- Investigated bacteriophage–bacteria interactions focusing on the role of bacterial multicellularity in phage resistance, phage isolation and characterisation
- Quantified bacterial growth and co-culture dynamics using growth curves and CFU counts; modelling of microbial population dynamics.

Tracking Behaviour of the Asian Elephant Population in the Kabini backwaters **May – July 2024**
Summer Research Internship under Prof. T. N. C. Vidya *JNCASR*
Animal Behaviour and Sociogenetics Lab

- Lagged associations in adult non-musth male elephants: Developed MATLAB (MATLAB version R204a) code to spatially cluster observations across temporal intervals and conducted social network analysis to investigate patterns of male-male associations in non-musth Asian elephants.
- Land-use mapping: Analysed temporal changes in the Kabini study area using Landsat-8 imagery, QGIS, and Google Earth; generated a Land Use/Land Cover (LULC) map using supervised classification with the QGIS Semi-Automatic Classification Plugin.

Understanding the role of *Drosophila melanogaster* in developmental biology **May – August 2023**
Summer Internship under Prof. Lolitika Mandal *IISER Mohali*
Drosophila Research Laboratory

- Fly management from larval to pupal stage, fly crosses, balancer chromosome study, imaginal disc dissection and mounting, nuclear staining (Hoechst and DAPI), spatial regulation using Dpp Gal4 system with various reporters like GFP and eyeless under regulation of UAS, temporal regulation using $Gal80^{ts}$, Cell lineage tracing and analysis using G-TRACE.

MEGA Plate under the supervision of Prof. N.G. Prasad & Dr. Shravan Mishra **September 2023**
***M**icrobial **E**volution **G**rowth **A**rena* *IISER Mohali*

- Our aim was to standardise the MEGA Plate Setup for demonstrating evolution on an ampicillin gradient
- Referenced and modified from the paper "Spatiotemporal microbial evolution on antibiotic landscapes" by Dr. Michael Baym et al.; used *Escherichia coli* DH5α transformed with pET-29b(+) plasmid with kan^R cassette

Workshops and Conferences

Ulm Biophysics Symposium 2026: Quantifying Biology Across Scales — **October 2026**
Attended talks on quantitative and physical approaches to biology covering super-resolution and single-molecule imaging, information theory in development and gene regulation, tissue mechanics and self-organization, organoids as a model system. Presented a poster titled "Diffusiophoresis in self-organizing systems in vitro" — *Ulm University, Germany*

6th VAAM Discussion Meeting: Microbial Cell Biology — **October 2026**
This meeting on bacterial and archael cell biology spanned topics in cell division, DNA organization membrane compartments and biomolecular condensates, secretion systems and motility, cell cycle regulation, and multicellular organization. Presented a poster titled "Diffusiophoresis in self-organizing systems in vitro" — *Marburg, Germany*

Spins, Games and Networks: From Data to Discovery in Complex systems — **December 2025**
The workshop at the The Institute of Mathematical Sciences (IMSc), Chennai focused on the theoretical and computational methods for modeling complex systems in the physical, biological and social sciences through pedagogical lectures, research talks and group projects. Themes included data science, machine learning, network science, game theory, nonlinear dynamics and statistical mechanics of classical spin systems, etc — *IMSc, Chennai*

Unifying Theories in High-Dimensional Biophysics — **July–Aug 2025**
Attended lectures at the interface of physics, biology and machine learning by eminent researchers held at the International Centre for Theoretical Sciences (ICTS), Bengaluru.Themes in the meeting covered areas across different scales of biology such as ecology, evolution, genetics, development, immunology, learning and neuroscience. I was among the few undergraduates to be selected for the conference — *ICTS, Bengaluru*

Science Media Workshop — **March 2025**
Attended a 5-day intensive Science Media Workshop organized by Dr. TV Venkateswaran featuring science communicator Gauhar Raza and documentary filmmaker Rakesh Andania — *IISER Mohali*

Other Projects

Modelling a Tale of Two Species: ODEs in Microbiome Research — **Spring semester, 2025**
IDC402: Nonlinear Dynamics and Chaos
Term paper for the Nonlinear Dynamics coursework: Explored the Lotka-Volterra and consumer-resource models to reproduce results from *Piccardi et al.'s* (2019) work on bacterial coculture dynamics

Exploring Waddington's Landscape in the Context of Cell-Fate Decisions — **Monsoon semester, 2025**
IDC401: Theoretical Biology
Term paper for the Theoretical Biology coursework: Explored Waddington's ideas in the context of cell fate decisions using geometry and dynamical systems theory. Reproduced results from existing literature on mathematical modelling of Waddington's genetic assimilation experiments landscapes *(Raju et al. 2023)*.

Awards and Scholarships

Academic Merit Certificate — **2024-2026**
Awarded 5 times by the institute for the best academic performance during the 4th to 8th semesters

Zonal Topper Award - Mimamsa, IISER Pune — **January, 2025**
Part of the Zonal Topper winning team for the states of Punjab and Haryana in the national-level annual science competition - Mimamsa, organized by IISER Pune for students at the undergraduate level

Summer Research Fellowship Program — **April 2024**
Awarded by the Indian Science Academies: Indian Academy of Sciences - Indian National Science Academy - National Academy of Sciences, India (IASc-INSA-NASI); (Did not accept the fellowship)

DST INSPIRE Scholarship — **2022-2027**
Scholarship for Higher Education (SHE) under the Department of Science and Technology, Government of India

Extracurricular Activities

Tutor- Genetics and evolution — **Spring semester 2025**
Conducting regular help sessions for junior batches for their Genetics and evolution course — *IISER Mohali*

Tutor- Genes and development — **Monsoon semester 2024**
Conducting regular help sessions for junior batches for their Genes and development course — *IISER Mohali*

Biology Department Representative, Member of SRC — **2024-2025**
Member of the Student Representative Council (SRC) — *IISER Mohali*

Core Organizing Committee, IISER Mohali's 18th Foundation Day — **September 2024**
Organized a science outreach event at IISER Mohali with scientific demonstrations, lectures, quizzes and competitions for 2000 school students across the tri-city — *IISER Mohali*

Co-Convenor, Biology Discussion Forum (BDF) — **2023–2024**
IISER Mohali's Biology Club — *IISER Mohali*

Member, IISER Mohali's Sports Contingent **2023–present**
Representing IISER Mohali in the Inter IISER Sports Meet as part of the Basketball, Football and Athletics Team *IISER Mohali*

Member, Manthan's Outreach Team **2023– present**
IISER Mohali's student-run magazine *IISER Mohali*

Volunteering Experience

Vigyan Jyoti Conclave **September 2024**
Volunteered to demonstrate Biology experiments to high school students as part of Vigyan Jyoti – a program initiated by the Department of Science & Technology in India to promote the participation of meritorious girls in STEM fields by providing support and resources from a high-school level *IISER Mohali*

Inter-IISER Cultural Meet (IICM) Volunteer **December 2023**
Volunteered for IICM '23 held at IISER Mohali *IISER Mohali*

Foundation Day Volunteer **September 2023**
Volunteered to demonstrate an experiment on bacterial evolution *IISER Mohali*

iGEM Volunteer **2023**
Volunteered for the 2023 iGEM Social Media and Outreach Team *IISER Mohali*